

READ: REINFORCEMENT-GUIDED CAPABILITY DISTILLATION FOR LARGE LANGUAGE MODELS

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ABSTRACT

Capability distillation specializes a smaller language model toward a specified capability using supervision from a larger teacher. Under a fixed distillation-token budget, an important question is how to allocate supervision across capabilities to improve the target most effectively. Our empirical study shows that single-capability distillation produces budget-dependent cross-capability changes and diminishing target gains, motivating allocation strategies that adapt to the student’s evolving state. We propose *ReAD*, a *Reinforcement-guided cApability Distillation* framework that combines a task-conditioned allocation prior, on-the-fly teacher supervision, and an uncertainty-aware contextual bandit. *ReAD* updates its allocation using measured capability changes, rewarding task-aligned gains and penalizing regressions on prioritized capabilities. Across eight target capabilities, two teacher–student pairs, and two token budgets, *ReAD* achieves higher mean on-target scores than the evaluated target-only baselines. Matched-generator comparisons show additional gains over static and greedy allocation schedules, while transfer diagnostics show smaller off-target declines. Our code is available at <https://anonymous.4open.science/r/ReAD-62B7>.

1 INTRODUCTION

Knowledge distillation (KD) (Hinton et al., 2015) for large language models (LLMs) (Zhao et al., 2023) has become an essential research direction for enhancing the accessibility and efficiency of Machine-Learning-as-a-Service (MLaaS) (Cai et al., 2024; Xu et al., 2024; Yang et al., 2024): *a small LLM learns to imitate a large teacher’s outputs and can then be deployed under limited computational or financial resources* (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Li et al., 2021; Liang et al., 2023). Recent studies show that capability distillation, a specialization of knowledge distillation that focuses supervision on a target capability (e.g., instruction-following, reasoning, mathematics, or coding), can substantially improve the student’s performance on downstream tasks that primarily depend on that capability, while achieving these gains at lower serving cost than the large teacher model (Magister et al., 2023; Shridhar et al., 2023; Xu et al., 2023a; Yue et al., 2024).

For a specified target capability, a natural distillation strategy is to devote the available supervision budget entirely to that capability (Taori et al., 2023; Chiang et al., 2023; Zhang et al., 2024). However, distillation updates also affect other capabilities (Zhong et al., 2024; Fang et al., 2025; Cloud et al., 2025), and the gains from additional target-specific supervision can diminish as training progresses. These observations motivate a budget-allocation question: *can adapting the mixture of capability-targeted supervision improve the target capability more effectively than devoting the entire budget to its own data?* To ground this question, we run a controlled empirical study: we repeatedly distill the student under several budgets, each time spending the entire budget on one of eight core capabilities (General Knowledge, Reasoning, Math, Code, Tool Use, Long-Context Understanding or LCU, Steerability, and Multilinguality), and evaluate it on all of them. This yields a budget-dependent capability transfer matrix whose diagonal holds on-target gains and whose off-diagonal entries show how performance is redistributed. Across budgets, we observe two consistent patterns: (i) *capability-specific distillation induces systematic, budget-dependent transfer to other capabilities rather than isolated improvements*, and (ii) *increasing the budget for a single target produces smaller additional target gains while making the average harm to non-target capabilities more pronounced*. These patterns motivate the allocation question but do not settle it: they show that the value of a

token depends on where the budget has already gone, not that mixing supervision must outperform target-only distillation, which our experiments test directly.

Building on these observed insights, we propose *ReAD*, a *Reinforcement-guided cApability Distillation* framework for large language models (Figure 2). *ReAD* first learns task-conditioned capability priorities, a simplex-constrained allocation prior over capabilities. It then spends the budget in short steps, distilling the student on teacher answers to prompts sampled from the current allocation and feeding the measured capability change to an uncertainty-aware contextual bandit, whose reward favors task-aligned gains and adds a penalty for regressions on prioritized capabilities. Online data mixing has been studied for pre-training and skill learning (Chen et al., 2023; Albalak et al., 2023; Chen et al., 2025); *ReAD* studies target-oriented allocation in teacher-supervised distillation, with rewards built from measured capability changes. Across eight target capabilities, two teacher-student pairs, and two budgets, *ReAD* achieves higher mean on-target scores than target-only baselines, improves over static and greedy allocation schedules with the generator held fixed, and shows smaller off-target declines than single-capability distillation.

To sum up, this paper makes the following key contributions:

- **Empirical observation.** In our models and budgets, single-capability distillation causes budget-dependent cross-capability transfer and diminishing target gains, motivating adaptive allocation.
- **Method.** *ReAD* allocates the budget stepwise, combining a task-conditioned prior, measured capability feedback, and uncertainty-aware exploration, with a local analysis of its mechanisms.
- **Experiments.** *ReAD* achieves higher mean on-target scores on eight target-capability distillation tasks, and matched-generator comparisons and transfer diagnostics show where gains come from.

2 PRELIMINARY

2.1 NOTATION AND FORMALIZATION

In this paper, we study budgeted capability distillation from a large teacher model T to a smaller student model S , with total training-token budget B . Let $\mathcal{C} = \{c_1, \dots, c_{|\mathcal{C}|}\}$ denote a set of measurable capabilities, here the eight core capabilities of Appendix Table 3, and let $s_k(M)$ be model M ’s benchmark score on capability c_k . A distillation strategy π specifies how data is generated, how the teacher is queried, and how tokens are allocated across capabilities, assigning budgets $\{b_k\}$ with $\sum_k b_k \leq B$, and producing a distilled student S_π .

Definition 2.1 (Capability distillation). Given a teacher model T , an initial student model S_0 , and capability-specific data distributions $\{\mathcal{D}_c\}_{c \in \mathcal{C}}$, capability distillation allocates a limited budget B across capabilities to improve task-relevant performance under shared representations by optimizing a weighted mixture of distillation objectives:

$$\min_{S, \mathbf{w}} \mathbb{E}_{x \sim \mathcal{D}_{\mathbf{w}}} [\ell(S(x), T(x))] \quad \text{s.t.} \quad \mathbf{w} \in \Delta_{|\mathcal{C}|}, \text{cost}(S; S_0) \leq B. \quad (1)$$

where $\mathcal{D}_{\mathbf{w}} := \sum_{c \in \mathcal{C}} w_c \mathcal{D}_c$, and $\text{cost}(S; S_0)$ measures the distillation token budget consumed to obtain S from S_0 . Here $\mathbf{w} = (w_c)_{c \in \mathcal{C}}$ lies in the probability simplex $\Delta_{|\mathcal{C}|} = \{\mathbf{w} \in \mathbb{R}^{|\mathcal{C}|} \mid w_c \geq 0, \sum_{c \in \mathcal{C}} w_c = 1\}$, which specifies the allocation of training effort across capabilities, and ℓ denotes a standard distillation loss, such as token-level cross-entropy or logit-matching between the student’s and the teacher’s output distributions.

Under shared representations, budget spent on one capability changes the others. We measure this cross-capability effect by distilling with one-hot allocations:

Definition 2.2 (Capability interdependence). Let S_0 be an initial student model, and let $S(\mathbf{w}, B)$ denote the student obtained after capability distillation under allocation $\mathbf{w} \in \Delta_{|\mathcal{C}|}$ and budget B . For each capability c_i , define the one-hot allocation vector $\mathbf{w}^{(i)} \in \Delta_{|\mathcal{C}|}$ by $w_i^{(i)} = 1$ and $w_j^{(i)} = 0$ for all $j \neq i$. The *capability interdependence* under capability distillation is characterized by the capability transfer matrix $\mathbf{T}(B) \in \mathbb{R}^{|\mathcal{C}| \times |\mathcal{C}|}$ with entries

$$\mathbf{T}_{ij}(B) := s_j(S(\mathbf{w}^{(i)}, B)) - s_j(S_0),$$



Figure 1: Exploratory study. (a–b) Cross-capability transfer under single-capability distillation: larger budgets sharpen target gains but expose stronger negative transfer. (c–d) Budget waste: extra tokens yield smaller target gains while increasing harmful spillover to non-target capabilities.

where $T_{ij}(B)$ quantifies the effect of distilling toward capability c_i on the performance of capability c_j . Non-zero off-diagonal entries indicate interdependence between capabilities under the shared representations of the student model.

2.2 EXPLORATORY STUDY AND PROBLEM FORMULATION

We distill Llama-3.3-70B-Instruct into Llama-3.1-8B-Instruct under fixed token budgets, one target capability at a time, and evaluate the student on all benchmarks to form the transfer matrix of Definition 2.2 (strategies and settings in Appendix B). All benchmark metrics are accuracies or success rates on a 0–100 scale, a capability score is the unweighted mean over its benchmarks, and every entry of the transfer matrix is an absolute difference in these points (Definition 2.2); no further rescaling is applied (Appendix B).

Observation 1: Distilling a specific capability redistributes performance across other capabilities in a budget-dependent manner. Figure 1(a–b) shows the transfer matrices at a small and a large budget. The off-diagonal mass is consistently non-zero, so optimizing one capability moves the others rather than leaving them unchanged, and the structure is budget-dependent: at small budgets the changes are weak and diffuse, whereas at large budgets the diagonal sharpens and negative off-diagonals appear for specific non-targets.

Observation 2: Capability distillation exhibits substantial budget waste due to diminishing returns and negative spillover. For the strongest-improving targets, the marginal target gain of 20M→80M is consistently larger than that of 80M→150M (*diminishing returns*, Figure 1c), while the drop on non-target capabilities grows with the target’s budget (*negative spillover*, Figure 1d). Beyond a capability-specific knee point, extra tokens thus buy little target benefit and amplify cross-capability degradation. Neither effect is new in isolation: off-target drops are a form of catastrophic forgetting (Kirkpatrick et al., 2017; Luo et al., 2023), and saturation is ordinary diminishing returns. What allocation can exploit is their *structure*: transfer is signed, source-specific, and budget-dependent, so a token’s marginal value depends on where the budget has already gone.

We formalize the allocation of a fixed budget as a sequential problem; in our experiments, the task τ is a specified target capability and U_τ is its on-target score:

Problem 2.3 (Budget allocation for target-capability distillation). Given a teacher model T , an initial student model S_0 , a downstream task τ , and a total distillation budget B , consider a sequential distillation process over steps $t = 1, \dots, T$ with cumulative cost not exceeding B . At each step t , a capability allocation vector $\mathbf{w}_t \in \Delta_{|C|}$ is selected and applied to update the student, yielding S_t . Let $U_\tau(\cdot)$ be the task-dependent utility function; our target is to determine an allocation policy $\{\mathbf{w}_t\}_{t=1}^T$ that maximizes the performance of the final student model:

$$\max_{\{\mathbf{w}_t\}_{t=1}^T} U_\tau(S_T), \text{ s.t. } \sum_{t=1}^T \text{cost}(S_{t-1} \rightarrow S_t) \leq B \quad (2)$$

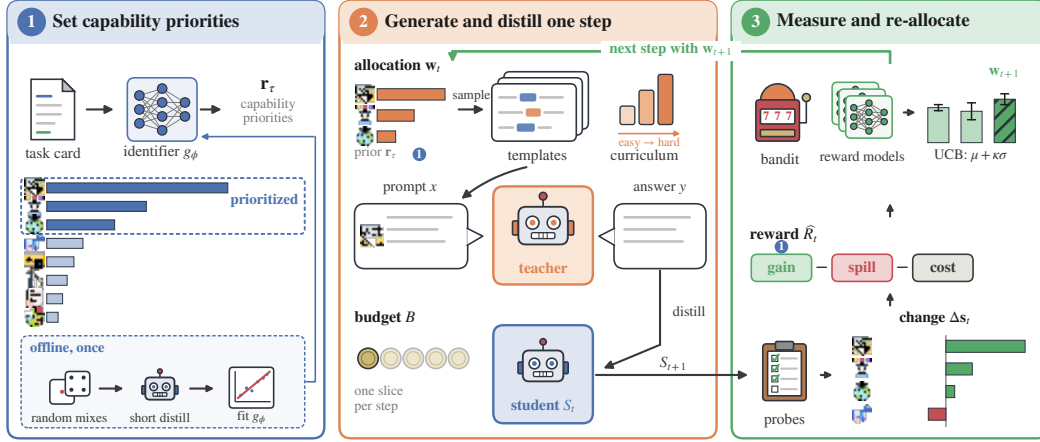


Figure 2: Overview of ReAD. (1) An identifier g_ϕ , pretrained once on low-budget probe interventions, maps a task card to the capability-priority vector $\mathbf{r}_\tau$, whose top- k entries form the prioritized set $\mathcal{C}_{\text{pri}}$. (2) At step t , capabilities are sampled from the allocation $\mathbf{w}_t$, prompts are instantiated from typed templates under a difficulty curriculum, and the student is distilled on the teacher’s answers. (3) The probe-profile change $\Delta \mathbf{s}_t$ gives the proxy reward, which trains a bootstrap ensemble whose mean μ and uncertainty σ select $\mathbf{w}_{t+1}$ by UCB; the loop repeats until B is spent. Values are illustrative.

3 METHODOLOGY

ReAD treats Problem 2.3 as state-dependent budget allocation over capabilities (Figure 2). It first sets task-conditioned capability priorities; then, at each step, it generates capability-targeted teacher supervision and uses an uncertainty-aware contextual bandit to choose the next token allocation. As the student profile changes, ReAD can shift budget away from saturated or high-spillover capabilities. ReAD is *off-policy* (Agarwal et al., 2024): the student learns from teacher completions, and on-policy losses (Gu et al., 2024; Ko et al., 2024) could replace this objective without changing the allocator.

3.1 TASK-CONDITIONED CAPABILITY PRIORITIES

Capability-priority vector. For each downstream task τ , ReAD builds a task card $\mathcal{D}_\tau^{\text{spec}}$ from non-test data (a short description, input/output format, evaluation target, and a few exemplars) and estimates from it a capability-priority vector $\mathbf{r}_\tau \in \Delta_{|C|}$, a simplex-constrained allocation prior in which $r_{\tau,c}$ is the priority of capability c for the task utility $U_\tau(\cdot)$. High-mass capabilities are prioritized in the candidate allocations, and regressions on them are penalized. It lives on the simplex because token allocations cannot be negative; negative sensitivities enter through signed capability changes and the regression penalty of Section 3.3.

Priority identifier and its supervision. A lightweight identifier g_ϕ (a small Transformer encoder over the task card, a two-layer MLP, and a softmax head) maps the card to $\mathbf{r}_\tau = g_\phi(\mathcal{D}_\tau^{\text{spec}})$. Since $\mathbf{r}_\tau$ is not observed, we supervise it through low-budget interventions in capability space. Writing $U_\tau(S) = F_\tau(\mathbf{s}(S))$ for the measured capability profile $\mathbf{s}(S) \in \mathbb{R}^{|C|}$, a small intervention that moves the profile by $\Delta \mathbf{s}$ changes the utility by

$$\Delta U_\tau = U_\tau(S') - U_\tau(S) \approx \nabla_{\mathbf{s}} F_\tau(\mathbf{s})^\top \Delta \mathbf{s} \quad (3)$$

(first-order expansion, Appendix C). This expansion motivates the training signal only: $\mathbf{r}_\tau$ is not an estimate of the true sensitivity but an allocation prior whose inner product with a capability change is fit to the observed utility change, as follows.

Offline pretraining. We build an intervention set $\mathcal{Z} = \{(\mathcal{D}_\tau^{\text{spec}}, \Delta \mathbf{s}_\tau^{(m)}, \Delta U_\tau^{(m)})\}_{\tau,m}$ from auxiliary tasks and non-test splits: for each task we sample sparse allocation vectors $\mathbf{w}^{(m)} \in \Delta_{|C|}$ (a random support with Dirichlet weights), run a short probe distillation with a small budget b_{probe} to obtain an intervened student $S^{(m)}$, and record $\Delta U_\tau^{(m)} = U_\tau(S^{(m)}) - U_\tau(S_0)$ and $\Delta \mathbf{s}_\tau^{(m)} = \mathbf{s}(S^{(m)}) - \mathbf{s}(S_0)$.

The identifier is pretrained to predict the utility change from the capability change,

$$\min_{\phi} \sum_{(\tau, m)} \left(\Delta U_{\tau}^{(m)} - g_{\phi}(\mathcal{D}_{\tau}^{\text{spec}})^{\top} \Delta \mathbf{s}_{\tau}^{(m)} \right)^2 + \lambda_{\text{ent}} \mathcal{H}(g_{\phi}(\mathcal{D}_{\tau}^{\text{spec}})), \quad (4)$$

where the entropy penalty discourages near-uniform priority vectors. At deployment, $\mathbf{r}_{\tau}$ is computed once per task and reused for the whole run; because the identifier is trained once and shared across tasks, it is not charged to the per-run distillation budget.

3.2 ON-THE-FLY CAPABILITY-TARGETED DATA GENERATION

Given $\mathbf{r}_{\tau}$, ReAD instantiates prompts on the fly from a capability-labeled template library, where each template specifies an instruction scaffold, typed slots filled from deterministic seeds, and output-format constraints. A deterministic difficulty score $d_c(x) \in [0, 1]$ averages capability-specific control factors (e.g., reasoning hops, required functions and tests, or schema rules; Eq. 10 in Appendix C) and splits prompts into easy, medium, and hard tertiles. At step t , ReAD samples capabilities according to $\mathbf{w}_t$, queries the teacher T for a completion y , and immediately distills on (x, y) . A curriculum moves the within-capability mix from easy toward hard examples without affecting the allocation, and per-template caps keep any single scaffold from dominating the generated data.

3.3 CONTEXTUAL BANDIT FOR CAPABILITY ALLOCATION

Decision loop. ReAD runs for $T_{\text{step}} = 20$ decision steps, with the bandit update, data resampling, and proxy refresh at every step (Section 5). At step t , the context summarizes the task priorities, current student profile, remaining budget, and recent allocation history:

$$\mathbf{x}_t = [\mathbf{r}_{\tau}; \mathbf{s}^{\text{probe}}(S_t); b_t; \boldsymbol{\rho}_t], \quad (5)$$

where $\boldsymbol{\rho}_t$ stores a short record of recent allocations and observed gains. The profile $\mathbf{s}^{\text{probe}}(S_t)$ is computed with a small fixed monitoring suite, while full held-out benchmark evaluation is used only for final reporting and coarse calibration.

Candidate allocations and distillation. At each step, ReAD selects an allocation $\mathbf{w}_t$ from a finite candidate set $\mathcal{A}(\tau)$ consisting of local perturbations around the previous allocation and sparse actions concentrated on the top- k capabilities under $\mathbf{r}_{\tau}$. Weights are chosen from a fixed grid and renormalized to sum to one. Given $\mathbf{w}_t$, ReAD samples capability-targeted data and updates the student by minimizing $\mathcal{L}_{\text{distill}}(\theta_t) = -\mathbb{E}_{(x, y)} \sum_{j=1}^{|y|} \log p_{\theta_t}(y_j \mid x, y_{<j})$ to produce S_{t+1} .

Proxy reward. After the update, ReAD measures the probe-profile change $\Delta \mathbf{s}_t^{\text{probe}} = \mathbf{s}^{\text{probe}}(S_{t+1}) - \mathbf{s}^{\text{probe}}(S_t)$ and forms

$$\hat{R}_t = \mathbf{r}_{\tau}^{\top} \Delta \mathbf{s}_t^{\text{probe}} - \beta \text{Spill}_t - \lambda \text{cost}_t, \quad \text{Spill}_t = \sum_{c \in \mathcal{C}_{\text{pri}}} r_{\tau, c} [-\Delta s_{t, c}^{\text{probe}}]_+ \quad (6)$$

Here $\mathcal{C}_{\text{pri}}$ contains the top- k (prioritized) capabilities under $\mathbf{r}_{\tau}$, and cost_t is the token budget consumed at the step. The first term rewards task-aligned capability gains and the second adds an extra penalty for regressions on prioritized capabilities, which already lower the first term. Because every step in our experiments consumes the same number of tokens, the third term is constant; the budget is enforced by the fixed total B and the update schedule.

Reward model and UCB allocation. We append $(\mathbf{x}_t, \mathbf{w}_t, \hat{R}_t)$ to a history buffer and train an ensemble of J two-layer MLP reward regressors $\{h_{\eta_j}\}_{j=1}^J$, each on a bootstrap resample of the 10% profiling split plus the logged transitions (3.2k–4.9k examples per task). For a candidate $\mathbf{w}$, $\mu(\mathbf{x}_t, \mathbf{w})$ and $\sigma(\mathbf{x}_t, \mathbf{w})$ denote the ensemble mean and standard deviation (Appendix C), and ReAD selects the next allocation with an upper-confidence rule,

$$\mathbf{w}_{t+1} = \arg \max_{\mathbf{w} \in \mathcal{A}(\tau)} \mu(\mathbf{x}_{t+1}, \mathbf{w}) + \kappa \sigma(\mathbf{x}_{t+1}, \mathbf{w}), \quad (7)$$

with $b_{t+1} = b_t - \text{cost}_t$. At coarse development checkpoints, we fit an affine calibration from cumulative proxy reward to observed utility change and reduce exploration if the development utility stagnates. Unlike static mixture-regression approaches such as RegMix (Liu et al., 2024), ReAD repeatedly re-estimates the best next allocation from the current student state and remaining budget.

4 THEORETICAL ANALYSIS

We give a local account of the three phenomena ReAD acts on: spillover, diminishing returns, and noisy reward estimates. The assumptions, full statements, and proofs are in Appendix D.

Local cross-capability transfer. Let θ_t be the student parameters after decision step t and $\Delta\theta_t = \theta_t - \theta_{t-1}$. Assume that (i) the expected update under allocation $\mathbf{w}_t$ is a mixture $\sum_c w_{t,c} d_c(\theta_{t-1})$ of capability-conditioned update directions, and (ii) each measured capability is locally a differentiable readout $\tilde{\mathbf{s}}(\theta)$. Then, by Proposition D.3, the change of each capability c at step t decomposes as

$$\Delta s_{t,c} \approx \sum_{c' \in \mathcal{C}} w_{t,c'} \Gamma_{c,c'}(\theta_{t-1}), \quad \Gamma_{c,c'}(\theta) := \nabla_{\theta} [\tilde{\mathbf{s}}(\theta)]_c^{\top} d_{c'}(\theta). \quad (8)$$

Γ is the local analogue of the empirical transfer matrix: an off-diagonal entry is nonzero whenever the update direction for c' is not orthogonal to the readout direction of c , which shared representations make common, so allocating budget to one capability alters others at every step.

Budget waste from diminishing returns. Let $b_c \geq 0$ be the tokens assigned to capability c , $\sum_c b_c \leq B$, and suppose task-aligned gains are concave in the budget, $\mathbb{E}[U_{\tau}(S) - U_{\tau}(S_0)] \approx \sum_c r_{\tau,c} G_{\tau,c}(b_c)$ with $G_{\tau,c}$ nondecreasing and concave. An optimal allocation equalizes weighted marginal gains across active capabilities (Proposition D.5), so moving a small budget from a single target $\bar{c}$ to another capability c' helps whenever $r_{\tau,c'} G'_{\tau,c'}(0) > r_{\tau,\bar{c}} G'_{\tau,\bar{c}}(b_{\bar{c}})$: once the target saturates, further tokens can be worth less than tokens spent elsewhere (Observation 2). This proxy ignores spillover, which ReAD’s reward in Eq. (6) penalizes separately.

Why uncertainty-aware allocation helps. Let $\bar{R}(\mathbf{x}, \mathbf{w})$ be the conditional expected proxy reward of allocation $\mathbf{w}$ in context $\mathbf{x}$. If the ensemble is calibrated, $|\mu - \bar{R}| \leq \kappa \sigma$ with high probability, the action chosen by Eq. (7) is within $2\kappa \sigma(\mathbf{x}_t, \mathbf{w}_t)$ of the best action in expected reward (Proposition D.7): the allocator can only be badly wrong where its own uncertainty is large, which is where the bonus directs exploration. This is a local, per-step statement, not a global guarantee for nonconvex fine-tuning.

5 EXPERIMENTAL SETTING

Models and budgets. We mainly distill Llama-3.3-70B-Instruct into Llama-3.1-8B-Instruct, and report Qwen2.5-72B-Instruct to Qwen2.5-14B-Instruct results in Appendix E. All methods use the same student checkpoint, training recipe, and 20M/150M distillation-token budgets.

Evaluation protocol. We evaluate eight capabilities using Appendix Table 3. For each benchmark, 10% of non-test data is used for profiling, **priority** inference, and allocation updates; the held-out split is reserved for final reporting. Generation uses temperature 0 with the same prompt template across methods. We report mean $\pm$ standard deviation over three seeds.

Baselines. For each capability, we build a held-out-filtered distillation pool from its benchmark prompts. For multi-dataset capabilities, we use balanced dataset-level sampling. The six main baselines combine three teacher signals—final response (Resp), chain-of-thought plus response (CoT), and token-level logits (Logit)—with two objectives, SFT and KD. Each baseline is one-hot at the capability level, spending the full budget on the target-capability pool. **We also compare against four allocation baselines that share ReAD’s generator, teacher, and budget and differ only in how $\mathbf{w}_t$ is chosen (Figure 4c):** *Uniform static* ($\mathbf{w}_t = \mathbf{1}/|\mathcal{C}|$), *Task-static mix* ($\mathbf{w}_t = \mathbf{r}_{\tau}$), *Greedy one-step* (re-selects $\mathbf{w}_t$ each step from the same context, including the remaining budget, but without the uncertainty bonus, $\kappa = 0$), and *Grid-searched* (best fixed allocation in $\mathcal{A}(\tau)$ on the profiling split). These baselines separate the allocation schedule from the rest of the pipeline. Finally, we include three published capability-specific distillation methods, adapted to our teacher–student pair and budget: step-by-step distillation (Hsieh et al., 2023) for reasoning, Equation-of-Thought Distillation (EoTD) (Zhu et al., 2024) for math, and CodePLAN (Sun et al., 2024) for code.

ReAD implementation. ReAD runs for 20 decision steps, with bandit updates, proxy refreshes, and data resampling every 1.0M tokens in the 20M setting and every 7.5M tokens in the 150M setting. Proxy refresh adds 3.12% wall-clock overhead, the full online loop adds 5.63%, and the reusable **priority** identifier costs 3.79 GPU-hours to train once. **Cost accounting.** B counts the teacher-generated tokens the student is trained on and is identical for all methods. Two ReAD-specific

Table 1: Target-capability performance on Llama under 20M- and 150M-token distillation budgets. Each baseline is distilled separately toward each target capability, and each column reports the corresponding on-target score. ReAD allocates supervision across capabilities for the corresponding target. Results are mean $\pm$ standard deviation over three seeds. Best results are bolded and second-best results are underlined in each column.

Method	General	Steer.	Reason.	Math	Code	Tool	LCU	Multi.
Teacher only	48.13 $\pm$ 0.35	89.98 $\pm$ 0.40	10.51 $\pm$ 0.12	48.34 $\pm$ 0.55	88.40 $\pm$ 0.70	31.90 $\pm$ 0.45	36.20 $\pm$ 0.40	91.10 $\pm$ 0.55
Student only	31.09 $\pm$ 0.40	49.22 $\pm$ 0.55	8.72 $\pm$ 0.15	15.56 $\pm$ 0.60	72.60 $\pm$ 0.95	25.83 $\pm$ 0.60	30.40 $\pm$ 0.45	68.90 $\pm$ 0.80
<i>Budget B = 20M</i>								
Resp-SFT	33.28 $\pm$ 0.47	52.84 $\pm$ 0.71	9.05 $\pm$ 0.18	17.12 $\pm$ 0.83	74.06 $\pm$ 1.09	26.92 $\pm$ 0.78	31.28 $\pm$ 0.51	70.02 $\pm$ 0.93
CoT-SFT	33.02 $\pm$ 0.52	52.16 $\pm$ 0.76	9.46 $\pm$ 0.14	18.58 $\pm$ 0.77	73.82 $\pm$ 1.02	26.81 $\pm$ 0.74	31.19 $\pm$ 0.57	69.78 $\pm$ 0.88
Logit-SFT	33.61 $\pm$ 0.44	53.04 $\pm$ 0.69	9.12 $\pm$ 0.17	17.63 $\pm$ 0.79	74.48 $\pm$ 1.15	27.03 $\pm$ 0.70	31.42 $\pm$ 0.53	70.21 $\pm$ 0.90
Resp-KD	33.83 $\pm$ 0.50	53.62 $\pm$ 0.73	9.18 $\pm$ 0.16	17.84 $\pm$ 0.74	74.23 $\pm$ 1.07	27.18 $\pm$ 0.76	31.58 $\pm$ 0.55	70.48 $\pm$ 0.95
CoT-KD	33.46 $\pm$ 0.46	52.98 $\pm$ 0.78	9.62 $\pm$ 0.13	19.34 $\pm$ 0.72	74.03 $\pm$ 1.10	27.06 $\pm$ 0.72	31.47 $\pm$ 0.58	70.08 $\pm$ 0.89
Logit-KD	34.02 $\pm$ 0.43	54.21 $\pm$ 0.66	9.25 $\pm$ 0.15	18.20 $\pm$ 0.76	74.62 $\pm$ 1.12	27.41 $\pm$ 0.69	31.72 $\pm$ 0.50	70.81 $\pm$ 0.91
ReAD	35.02 $\pm$ 0.41	58.03 $\pm$ 0.61	10.01 $\pm$ 0.12	21.06 $\pm$ 0.68	77.06 $\pm$ 0.97	29.82 $\pm$ 0.64	33.58 $\pm$ 0.49	73.04 $\pm$ 0.84
<i>Budget B = 150M</i>								
Resp-SFT	38.41 $\pm$ 0.58	68.73 $\pm$ 0.94	9.55 $\pm$ 0.14	28.72 $\pm$ 1.08	79.18 $\pm$ 1.26	28.53 $\pm$ 0.82	32.69 $\pm$ 0.73	78.23 $\pm$ 1.12
CoT-SFT	38.02 $\pm$ 0.61	67.92 $\pm$ 0.89	10.14 $\pm$ 0.12	31.58 $\pm$ 0.97	78.86 $\pm$ 1.33	28.41 $\pm$ 0.79	32.83 $\pm$ 0.68	77.77 $\pm$ 1.06
Logit-SFT	39.03 $\pm$ 0.56	69.38 $\pm$ 0.86	9.70 $\pm$ 0.13	29.61 $\pm$ 1.02	79.69 $\pm$ 1.21	28.71 $\pm$ 0.77	33.01 $\pm$ 0.70	78.63 $\pm$ 1.09
Resp-KD	39.62 $\pm$ 0.53	70.98 $\pm$ 0.84	9.82 $\pm$ 0.12	30.21 $\pm$ 0.98	80.09 $\pm$ 1.18	29.09 $\pm$ 0.74	33.42 $\pm$ 0.66	79.18 $\pm$ 1.03
CoT-KD	39.12 $\pm$ 0.55	69.84 $\pm$ 0.88	10.26 $\pm$ 0.11	33.02 $\pm$ 0.92	79.81 $\pm$ 1.24	29.01 $\pm$ 0.71	33.19 $\pm$ 0.69	78.71 $\pm$ 1.01
Logit-KD	40.18 $\pm$ 0.51	72.63 $\pm$ 0.79	9.95 $\pm$ 0.12	31.36 $\pm$ 0.95	80.57 $\pm$ 1.16	29.48 $\pm$ 0.69	33.73 $\pm$ 0.64	79.97 $\pm$ 0.97
ReAD	41.80 $\pm$ 0.45	78.20 $\pm$ 0.70	10.40 $\pm$ 0.10	35.20 $\pm$ 0.80	83.90 $\pm$ 1.00	31.30 $\pm$ 0.55	35.60 $\pm$ 0.55	84.10 $\pm$ 0.90

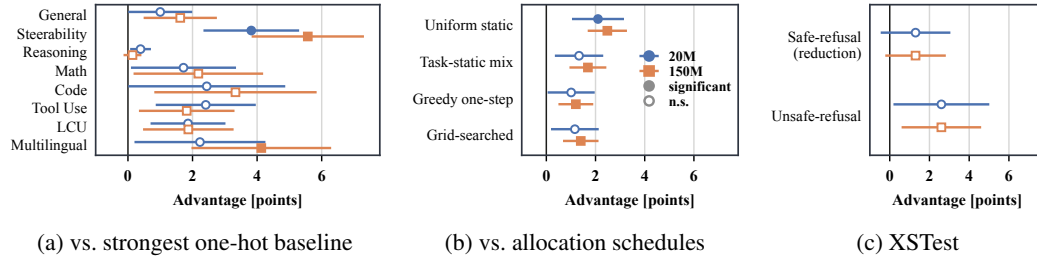


Figure 3: ReAD’s advantage with unadjusted 95% Welch intervals from the reported means and standard deviations (three seeds; blue circles 20M, orange squares 150M). Filled markers stay significant after Holm correction per panel and budget; safe-refusal is shown as a reduction, so positive values favor ReAD in all panels.

costs fall outside B and are reported separately rather than folded into it: the identifier is trained once from auxiliary tasks and reused for every task and budget, so its 3.79 GPU-hours are amortized, and the fixed probe suite is the 3.12% overhead above.

6 EXPERIMENTS

We study four questions: **RQ1** on-target performance across separate target-capability distillation tasks under the same budget; **RQ2** whether the allocation schedule matters when the generator is held fixed; **RQ3** budget efficiency and the trade-off between target gain and spillover; and **RQ4** transfer beyond capability-aligned benchmarks. Figure 3 gives a statistical view of every comparison.

6.1 RQ1: ON-TARGET PERFORMANCE UNDER A FIXED BUDGET

Table 1 reports on-target performance across separate distillation tasks: for each target capability, every baseline is distilled separately toward it, ReAD allocates supervision across capabilities for the same target, and all methods are scored on that target. ReAD achieves the highest mean on-target score for all eight targets at both budgets. The largest margins are on Steerability at both budgets, followed by Code and Tool Use at 20M and by Multilingual and Code at 150M. Because the baselines also differ from ReAD in how supervision is generated, this comparison measures the effect of the whole method; RQ2 compares allocation schedules under the same generator. The Qwen results

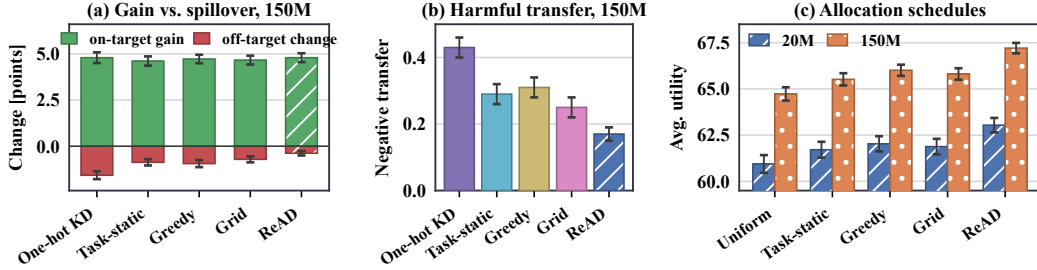


Figure 4: Budget-allocation diagnostics (mean $\pm$ std, three seeds; values in Appendix Table 8). (a) On-target gain (up), off-target change (down), and (b) negative transfer at 150M; (c) allocation schedules under a matched generator.

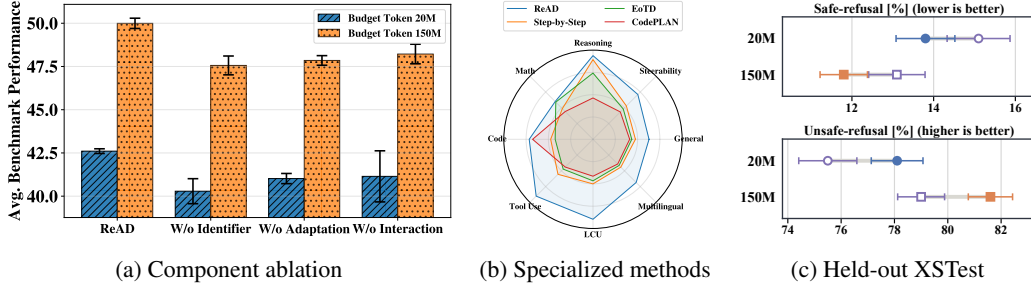


Figure 5: (a) Component ablation (average score over eight capabilities). (b) ReAD vs. step-by-step distillation, EoTD, and CodePLAN at 150M (fraction of the teacher–student gap closed). (c) XSTest: ReAD (filled) vs. the best SFT/KD baseline (hollow purple); blue 20M, orange 150M.

in Appendix E show the same ordering, suggesting that the improvement is not tied to a single teacher–student pair. With three seeds, all sixteen differences are positive and all but Reasoning at 150M are significant before correction; after Holm correction over the eight capabilities, Steerability at both budgets and Multilingual at 150M remain significant (Figure 3).

When the margin is large. The margin grows with the teacher–student headroom (Spearman $\rho = 0.74$ at 150M, $p = 0.04$): it is largest on Steerability and Multilingual (teacher ahead by over 20 points) and smallest on Reasoning (1.8 points). ReAD’s gain over the student is $1.9\times$ that of the strongest baseline at 20M and $1.3\times$ at 150M, so the relative advantage peaks when tokens are scarce.

Against the three capability-specific methods at 150M (Figure 5b), which concentrate gains on their own target as Section 2.2 predicts, ReAD distilled toward the same target is never worse on it.

6.2 RQ2: ALLOCATION SCHEDULES UNDER A MATCHED GENERATOR

Figure 4c compares ReAD with the four allocation baselines of Section 5, which share its generator, teacher, and token budget and differ only in how w_t is chosen. Static schedules keep one allocation for the whole run; Greedy one-step re-selects it every step from the same context, including the remaining budget, but without the uncertainty bonus. ReAD outperforms all four, with larger margins at 150M. The margins (+1.0/+1.2 over Greedy one-step and +1.2/+1.4 over Grid-searched at 20M/150M; seed std 0.28–0.42) are smaller than over the one-hot baselines because every schedule already mixes capabilities. Greedy one-step is only 0.15–0.20 points above the best static schedule, so in this comparison most of ReAD’s gain comes from uncertainty-aware selection on top of dynamic updates rather than from re-estimating the mix alone. All four gains remain significant after Holm correction at 150M, and the gain over Uniform static also at 20M (Figure 3).

Component ablation. Figure 5a changes one component at a time: *W/o Identifier* replaces the task-guided allocation with a uniform one (a component variant, not an isolated ablation of g_ϕ), *W/o Adaptation* fixes the allocation, and *W/o Interaction* removes the extra regression penalty. The first two cause the largest drops, and the regression penalty adds a smaller further gain.

Table 2: Interpolation-based estimates of the tokens the strongest one-hot baseline needs to match ReAD’s 20M score, from its 20M and 150M scores in Table 1 (ratio to 20M).

	General	Steer.	Reason.	Math	Code	Tool	LCU	Multi.
ReAD @20M	35.02	58.03	10.01	21.06	77.06	29.82	33.58	73.04
Strongest baseline	Logit-KD	Logit-KD	CoT-KD	CoT-KD	Logit-KD	Logit-KD	Logit-KD	Logit-KD
@20M / @150M	34.02/40.18	54.21/72.63	9.62/10.26	19.34/33.02	74.62/80.57	27.41/29.48	31.72/33.73	70.81/79.97
Log-linear interp.	28M (1.4×)	30M (1.5×)	68M (3.4×)	26M (1.3×)	46M (2.3×)	> 150M	129M (6.5×)	33M (1.6×)
Linear interp.	41M (2.1×)	47M (2.3×)	99M (5.0×)	36M (1.8×)	73M (3.7×)	> 150M	140M (7.0×)	52M (2.6×)

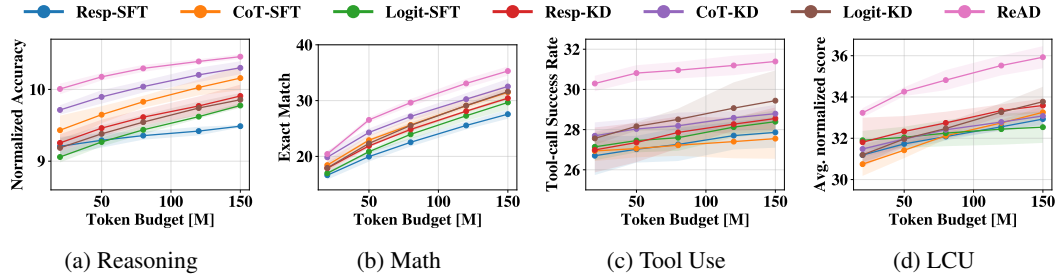


Figure 6: Performance versus token budget for four bottleneck capabilities. Curves show mean $\pm$ standard deviation over three seeds.

6.3 RQ3: BUDGET EFFICIENCY AND SPILLOVER

Figure 6 reports budget-scaling curves for Reasoning, Math, Tool Use, and LCU. ReAD achieves higher performance at **every** budget and **scales most steeply on Math and LCU**; on Reasoning and Tool Use the baselines narrow the gap by 150M as all methods approach the teacher. ReAD also gains the most per additional 10M tokens on six of the eight capabilities (Appendix Table 6). From only the two measured budgets, Table 2 estimates that matching ReAD’s 20M score would take the strongest one-hot baseline 1.3–6.5 $\times$ as many tokens (median about 2 $\times$) under log-linear and 1.8–7.0 $\times$ (median about 3 $\times$) under linear interpolation, and Tool Use is not matched even at 150M.

Figure 4a–b shows **the trade-off between target gain and spillover at 150M**. Single-capability KD achieves strong target gains but causes larger off-target degradation, while static mixing reduces spillover at the cost of weaker target improvement. **ReAD matches the on-target gain of single-capability KD with the smallest off-target decline and negative-transfer rate of all methods.**

6.4 RQ4: TRANSFER BEYOND ALIGNED BENCHMARKS

On held-out XSTest (Röttger et al., 2024), with the same pipeline and no new capability head, ReAD lowers safe-refusal and raises unsafe-refusal scores over the best SFT/KD baseline at both budgets (Figure 5c, Appendix Table 7). The four differences agree in direction, but none survives Holm correction with three seeds (Figure 3), so this is preliminary evidence of transfer beyond the eight capabilities rather than a firm result.

7 CONCLUSIONS

We study how to allocate a fixed distillation budget across capability-targeted supervision to improve a specified target capability. ReAD combines a task-conditioned allocation prior, on-the-fly teacher supervision, and an uncertainty-aware contextual bandit. Under the same budget, it achieves higher on-target scores than target-only baselines, improves over static and greedy schedules with a matched generator, and shows smaller off-target declines than single-capability distillation. Two caveats qualify these results: the proxy reward comes from a small probe suite and may deviate from true task utility, and the efficiency figures are interpolation-based estimates from two budgets. Validating the proxy against downstream utility and measuring more budget points are natural next steps.

AI USE STATEMENT

In this work, we used generative AI tools to assist with implementing parts of the experimental code (e.g., data processing, evaluation, and plotting scripts) and to improve the writing of the manuscript (grammar, wording, and formatting). We have not used generative AI tools to generate synthetic datasets, develop the theoretical model or conceptual framework, formulate or prove mathematical claims, propose or refine hypotheses, design the research methodology or experiments, or interpret results; the remaining tasks with required disclosure (e.g., translation and qualitative or thematic data analysis) are not applicable to this work. We have reviewed all AI-assisted work: AI-assisted code was checked and tested by the authors before use, and all AI-edited text was verified by the authors against the intended content. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work does not involve human subjects, crowdsourcing, or personally identifiable data. All experiments use publicly available pretrained models and public benchmarks, and evaluation is fully automated. ReAD is a method for allocating a fixed distillation budget across capabilities; it does not introduce new datasets or release new model weights. As with any distillation method, distilled students may inherit biases or unsafe behaviors from the teacher and the training data, and more efficient distillation may lower the cost of producing specialized models with dual-use capabilities. We discuss these risks and the corresponding mitigations (task-specific safety evaluation, bias and robustness audits, and monitoring of off-target capability changes) in Appendix G.

REPRODUCIBILITY STATEMENT

We have made efforts to ensure that our results are reproducible. Section 5 specifies the teacher-student model pairs, token budgets, evaluation protocol, baselines, and the ReAD update schedule, and Section 3 describes each component of ReAD. Appendix A lists the benchmarks, metrics, and data splits used for each capability, Appendix B details the setting of the exploratory study, and Appendix E reports additional experiments and ablations. The assumptions and full derivations for the theoretical results in Section 4 are given in Appendix D. All reported numbers are means and standard deviations over three seeds. An anonymized repository with our code, configuration files, and instructions for reproducing the main results is available at <https://anonymous.4open.science/r/ReAD-62B7> (anonymized for review).

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TECHNICAL APPENDICES AND SUPPLEMENTARY MATERIAL

A BENCHMARK DATASETS

Table 3 summarizes the benchmark suite used to evaluate each capability. For capabilities associated with multiple benchmarks, we report the average score across the listed benchmarks.

Table 3: Benchmark suite for evaluating LLM capabilities.

Capability	Benchmarks	Metric
General	MMLU-Pro, BIG	Accuracy
Steerability	IFEval, FollowEval	Instruction-following accuracy
Reasoning	GPQA, ARC-Challenge	Normalized accuracy
Math	MATH Hard, GSM8K	Exact match
Code	HumanEval, MBPP	Pass@1
Tool Use	BFCL V4, ToolBench	Tool-call success rate
LCU	LongBench	Average normalized score
Multilingual	XCOPA, MGSM	Macro accuracy

- **MMLU-Pro** (Wang et al., 2024) tests broad knowledge and problem-solving across many academic subjects using multiple-choice questions.
- **BIG-bench (BIG)** (Suzgun et al., 2022) is a large collection of diverse tasks designed to probe general reasoning and knowledge beyond standard exams.
- **IFEval** (Zhou et al., 2023) measures instruction following by checking whether model outputs satisfy explicit, verifiable constraints in the prompt.
- **FollowEval** (Jing et al., 2023) evaluates how well a model follows user instructions under varied formats and constraint types, complementing IFEval with broader instruction patterns.
- **GPQA** (Rein et al., 2024) is a graduate-level question answering benchmark that stresses hard scientific reasoning with high-quality, expert-written questions.
- **ARC-Challenge** (Clark et al., 2018) contains difficult grade-school science questions that require multi-step reasoning and commonsense beyond keyword matching.
- **MATH Hard** (Hendrycks et al., 2021) is a challenging subset of competition-style math problems and is scored by exact final-answer match.
- **GSM8K** (Cobbe et al., 2021) focuses on grade-school math word problems that require intermediate reasoning steps, also scored by exact final-answer match.
- **HumanEval** (Chen et al., 2021) evaluates code generation via functional correctness on hand-written programming problems, reported as Pass@1.
- **MBPP** (Austin et al., 2021) is a Python programming benchmark with short programming tasks and unit tests, also evaluated by Pass@1.
- **BFCL V4** (Yan et al., 2024) measures tool-use ability by checking whether the model can produce correct function calls under realistic tool specifications.
- **ToolBench** (Xu et al., 2023b) evaluates tool use in more complex, multi-tool scenarios and is scored by tool-call success rate.
- **LongBench** (Bai et al., 2024) evaluates long-context understanding across multiple long-input tasks and reports an average normalized score.
- **XCOPA** (Ponti et al., 2020) tests multilingual causal commonsense reasoning across languages and is reported using macro accuracy.
- **MGSM** (Shi et al., 2022) is a multilingual version of grade-school math and is evaluated by macro accuracy across languages.

B EMPIRICAL STUDY SETTING

We study capability interactions under a fixed distillation budget by distilling a student model toward one target capability at a time and then evaluating the resulting student on *all* capabilities in Table 3. For each capability, we build a capability-specific distillation pool from the corresponding benchmark prompts, excluding any held-out evaluation examples to prevent leakage. We repeat this process under multiple token budgets (e.g., 20M, 80M, 150M tokens) to form a budget-dependent capability transfer matrix, where each entry is the score change on a destination capability when distilling toward a source capability at budget B . We use deterministic decoding for generation benchmarks (temperature = 0) with a fixed prompt template across methods, and we compute each capability score as the average over its associated benchmarks in the table. We instantiate a simple but representative set of capability distillation baselines by combining three supervision forms that expose different levels of teacher knowledge—*Resp* (final answer only), *CoT* (rationale plus final answer), and *Logit* (token-level soft targets)—with two standard objectives, *SFT* and *KD*, yielding six baselines in total.

Scoring and normalization. Each benchmark in Table 3 is scored with its standard metric (accuracy, exact match, pass@1, or tool-call success rate) expressed in percentage points; LongBench reports its own per-task normalized score on the same 0–100 scale. A capability score is the unweighted mean of its benchmark scores, and a transfer-matrix entry $\mathbf{T}_{ij}(B)$ is the absolute difference in capability score between the distilled and the initial student. No min–max rescaling or gap-closure normalization is applied to the reported tables or transfer matrices, so a change of one point is one percentage point of the underlying metric, and small absolute changes remain small in the figures.

C METHOD DETAILS

The following details were moved here from Section 3.

Local utility sensitivity. With $U_\tau(S) = F_\tau(\mathbf{s}(S))$, a small intervention that changes the profile from $\mathbf{s}$ to $\mathbf{s} + \Delta\mathbf{s}$ changes the utility according to the first-order expansion

$$F_\tau(\mathbf{s} + \Delta\mathbf{s}) = F_\tau(\mathbf{s}) + \nabla_{\mathbf{s}} F_\tau(\mathbf{s})^\top \Delta\mathbf{s} + o(\|\Delta\mathbf{s}\|), \quad (9)$$

which yields the local approximation in Eq. (3). This does not assume that the true sensitivity has no negative coordinates. The expansion only motivates the training signal of Eq. (4): the priority vector $\mathbf{r}_\tau$ is a nonnegative, simplex-constrained allocation prior over capabilities, not an estimate of the sensitivity itself; negative sensitivities enter through the signed capability changes in the reward.

Difficulty scoring. For prompt x in capability c , we compute a deterministic difficulty score

$$d_c(x) = \frac{1}{|\mathcal{F}_c|} \sum_{f \in \mathcal{F}_c} \frac{f(x) - \min_{x' \in \mathcal{Q}_c} f(x')}{\max_{x' \in \mathcal{Q}_c} f(x') - \min_{x' \in \mathcal{Q}_c} f(x') + \epsilon}, \quad (10)$$

where $\mathcal{F}_c$ is the set of capability-specific control factors and $\mathcal{Q}_c$ is a calibration prompt pool for capability c . For example, reasoning templates use factors such as the number of constraints, reasoning hops, and symbolic complexity; code templates use the number of required functions and tests; steerability templates use the number of rules and schema fields. We split each capability’s calibration pool into easy, medium, and hard buckets by tertiles of $d_c(x)$.

Ensemble mean and uncertainty. For a candidate allocation $\mathbf{w}$, the J reward regressors give the ensemble mean and the ensemble standard deviation

$$\mu(\mathbf{x}_t, \mathbf{w}) = \frac{1}{J} \sum_{j=1}^J h_{\eta_j}(\mathbf{x}_t, \mathbf{w}), \quad \sigma(\mathbf{x}_t, \mathbf{w}) = \sqrt{\frac{1}{J-1} \sum_{j=1}^J (h_{\eta_j}(\mathbf{x}_t, \mathbf{w}) - \mu(\mathbf{x}_t, \mathbf{w}))^2}. \quad (11)$$

D ADDITIONAL THEORETICAL ANALYSIS

This appendix states the assumptions and results summarized in Section 4. The first two parts were moved here from the main text for space.

D.1 LOCAL CROSS-CAPABILITY TRANSFER

Let $\theta_t \in \mathbb{R}^p$ denote the parameters of the student S_t , and let $\Delta\theta_t = \theta_t - \theta_{t-1}$. We analyze one decision step, which is the regime in which ReAD re-estimates its allocation.

Assumption D.1 (Mixture-structured update). At interval t , given an allocation vector $\mathbf{w}_t \in \Delta_{|C|}$, the expected update admits a local mixture approximation $\mathbb{E}[\Delta\theta_t \mid \mathbf{w}_t, \theta_{t-1}] \approx \sum_{c \in C} w_{t,c} d_c(\theta_{t-1})$, where $d_c(\theta)$ is the capability-conditioned local update direction.

Assumption D.2 (Local capability readout). The measured capability profile $\mathbf{s}(S)$ is represented locally as a differentiable function $\tilde{\mathbf{s}}(\theta)$. Near θ_{t-1} ,

$$\Delta s_{t,c} = [\mathbf{s}(S_t)]_c - [\mathbf{s}(S_{t-1})]_c \approx \nabla_{\theta} [\tilde{\mathbf{s}}(\theta_{t-1})]_c^{\top} \Delta\theta_t. \quad (12)$$

Proposition D.3 (Local transfer decomposition). *Under Assumptions D.1 and D.2, the expected change in capability c satisfies Eq. (8), i.e., $\Delta s_{t,c} \approx \sum_{c' \in C} w_{t,c'} \Gamma_{c,c'}(\theta_{t-1})$ with $\Gamma_{c,c'}(\theta) := \nabla_{\theta} [\tilde{\mathbf{s}}(\theta)]_c^{\top} d_{c'}(\theta)$.*

Proof. Substituting the mixture update of Assumption D.1 into the local readout of Assumption D.2 yields, for every capability c and step t ,

$$\Delta s_{t,c} \approx \nabla_{\theta} [\tilde{\mathbf{s}}(\theta_{t-1})]_c^{\top} \sum_{c'} w_{t,c'} d_{c'}(\theta_{t-1}) = \sum_{c'} w_{t,c'} \Gamma_{c,c'}(\theta_{t-1}).$$

□

Interpretation. Γ is the local analogue of the empirical transfer matrix: diagonal terms are intended gains, and off-diagonal terms are nonzero whenever the update direction for c' is not orthogonal to the readout direction of c , which is expected under shared representations. This is why allocating budget to one capability alters others at the decision-step level.

D.2 BUDGET WASTE FROM DIMINISHING RETURNS

The previous subsection explains spillover. We now isolate diminishing returns with a simpler cumulative-budget proxy. Let $b_c \geq 0$ be the total number of distillation tokens assigned to capability c , with $\sum_c b_c \leq B$, and let $\mathbf{r}_{\tau} \in \Delta_{|C|}$ be the task requirement vector.

Assumption D.4 (Concave task-aligned gains). For task τ and capability c , there is a nondecreasing concave gain function $G_{\tau,c} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that, **for every allocation with $\sum_{c \in C} b_c \leq B$, $\mathbb{E}[U_{\tau}(S) - U_{\tau}(S_0)] \approx \sum_{c \in C} r_{\tau,c} G_{\tau,c}(b_c)$, i.e., task-aligned gains add up across capabilities.**

Consider a restricted distillation strategy that allocates budget only to a subset $\mathcal{K} \subseteq C$:

$$\max_{\{b_c \geq 0\}_{c \in \mathcal{K}}} \sum_{c \in \mathcal{K}} r_{\tau,c} G_{\tau,c}(b_c) \quad \text{s.t.} \quad \sum_{c \in \mathcal{K}} b_c \leq B. \quad (13)$$

Proposition D.5 (Weighted marginal gain criterion). *Any optimum $\{b_c^*\}_{c \in \mathcal{K}}$ of Eq. (13) equalizes weighted marginal gains among active capabilities: there exists $\nu^* \geq 0$ such that*

$$r_{\tau,c} G'_{\tau,c}(b_c^*) \leq \nu^* \quad \forall c \in \mathcal{K}, \quad r_{\tau,c} G'_{\tau,c}(b_c^*) = \nu^* \quad \text{if } b_c^* > 0. \quad (14)$$

Moreover, in the single-capability case targeting $\bar{c}$, shifting an infinitesimal budget from $\bar{c}$ to another task-relevant capability c' improves the proxy objective whenever $r_{\tau,c'} G'_{\tau,c'}(0) > r_{\tau,\bar{c}} G'_{\tau,\bar{c}}(b_{\bar{c}})$.

Proof. The objective is a nonnegative combination of concave functions and the feasible set is convex, so the Karush–Kuhn–Tucker conditions are necessary and sufficient. With ν^* the multiplier of the budget constraint and $\xi_c \geq 0$ those of $b_c \geq 0$, stationarity gives $r_{\tau,c} G'_{\tau,c}(b_c^*) = \nu^* - \xi_c \leq \nu^*$, and complementary slackness ($\xi_c b_c^* = 0$) gives equality whenever $b_c^* > 0$. For the single-target case, moving ϵ tokens from $\bar{c}$ to c' changes the objective by $\epsilon [r_{\tau,c'} G'_{\tau,c'}(0) - r_{\tau,\bar{c}} G'_{\tau,\bar{c}}(b_{\bar{c}})] + o(\epsilon)$, which is positive under the stated condition. □

Implication for ReAD. This formalizes budget waste: once the target saturates, further tokens have lower task-aligned marginal value than tokens assigned elsewhere. ReAD’s reward in Eq. (6) is richer than this proxy because it also penalizes observed spillover, so the allocator optimizes a step-level reward that accounts for both diminishing returns and cross-capability degradation.

D.3 UNCERTAINTY-AWARE UCB ALLOCATION

Here, we show that action-level regret is controlled by predictive uncertainty. At interval t , ReAD summarizes the priority vector, the current capability profile, and the remaining budget in the bandit context that the reward regressors take as input

$$\mathbf{x}_t := [\mathbf{r}_\tau; \mathbf{s}(S_t); b_t], \quad (15)$$

where $\mathbf{r}_\tau \in \Delta$ is the **capability-priority** vector, $\mathbf{s}(S_t) \in \mathbb{R}^{|C|}$ is the probe-defined capability profile of the current student, and b_t is the remaining budget. Let $\mathcal{A}(\tau)$ be the discrete candidate action set of allocation vectors. Define the conditional expected proxy reward

$$\bar{R}(\mathbf{x}, \mathbf{w}) := \mathbb{E}[\hat{R}_t \mid \mathbf{x}_t = \mathbf{x}, \mathbf{w}_t = \mathbf{w}], \quad (16)$$

where $\hat{R}_t$ is the proxy reward used as shown in Eq. (6).

Assumption D.6 (Calibrated uncertainty). For a confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, for all encountered $(\mathbf{x}_t, \mathbf{w})$,

$$|\mu(\mathbf{x}_t, \mathbf{w}) - \bar{R}(\mathbf{x}_t, \mathbf{w})| \leq \kappa \sigma(\mathbf{x}_t, \mathbf{w}), \quad (17)$$

where μ and σ are the ensemble mean and standard deviation defined from the reward regressors, and κ is the exploration coefficient of Eq. (7).

Since ReAD selects allocations by the UCB rule in Eq. (7), we can compare its chosen action $\mathbf{w}_t$ to the optimal action $\mathbf{w}_t^*$ under the same context and bound the resulting gap in conditional expected reward using the calibrated uncertainty condition in Assumption D.6, which leads to the following instantaneous regret bound **on the chosen action**.

Proposition D.7 (Instantaneous regret bound). *Let $\mathbf{w}_t^* \in \arg \max_{\mathbf{w} \in \mathcal{A}(\tau)} \bar{R}(\mathbf{x}_t, \mathbf{w})$. Under Assumption D.6, the action chosen by Eq. (7) satisfies*

$$\bar{R}(\mathbf{x}_t, \mathbf{w}_t^*) - \bar{R}(\mathbf{x}_t, \mathbf{w}_t) \leq 2\kappa \sigma(\mathbf{x}_t, \mathbf{w}_t), \quad (18)$$

and consequently, **summing over the T decision steps of the run**,

$$\sum_{t=1}^T \left(\bar{R}(\mathbf{x}_t, \mathbf{w}_t^*) - \bar{R}(\mathbf{x}_t, \mathbf{w}_t) \right) \leq 2\kappa \sum_{t=1}^T \sigma(\mathbf{x}_t, \mathbf{w}_t). \quad (19)$$

Proof sketch. By calibration, the true reward of any action is at most its upper confidence score $\mu + \kappa\sigma$, and the selected action maximizes this score; applying the calibration inequality once more to the selected action yields Eq. (18), and summing over steps gives Eq. (19). This result does not imply a global optimum for nonconvex fine-tuning; it justifies the local ranking rule used by ReAD at each budget step rather than a global guarantee. $\square$

E ADDITIONAL EXPERIMENTS

(The 150M Llama table was merged into Table 1 in the main text.)

Additional evidence showing that ReAD improves capability distillation under a fixed budget. We compare ReAD with six capability distillation baselines. Table 1 (main text) reports the Llama results at $B=20M$ and $B=150M$, and Tables 4 and 5 report the corresponding Qwen results. In both budgets and for both model families, ReAD achieves the highest mean on-target score on every target capability. The largest margin is on Steerability and the smallest on Reasoning in all four settings, in line with the headroom analysis of Section 6.1. Scaling the budget from 20M to 150M strengthens all methods; ReAD’s average margin over the strongest baseline grows for Llama (+2.0 to +2.6 points) and narrows slightly for Qwen (+1.8 to +1.6 points), whose stronger student leaves less headroom. Qwen starts from a stronger student and teacher on every capability and reaches higher absolute scores after distillation, but the strongest baseline for each capability and the top rank of ReAD are unchanged, suggesting that ReAD’s advantage is not tied to a specific LLM family. Figure 7 visualizes the same Llama numbers as absolute gains over the initial student, next to the best one-hot baseline of each capability, which makes the per-capability margins directly comparable.

Table 4: Target-specific distillation performance under a 20M-token budget (Qwen). ReAD outperforms all six one-hot baselines on every target.

Method	General	Steerability	Reasoning	Math	Code	Tool Use	LCU	Multilingual
Teacher only	55.27 $\pm$ 0.46	92.36 $\pm$ 0.52	14.28 $\pm$ 0.18	55.74 $\pm$ 0.78	90.12 $\pm$ 0.88	36.08 $\pm$ 0.63	40.23 $\pm$ 0.57	92.34 $\pm$ 0.66
Student only	40.07 $\pm$ 0.61	70.18 $\pm$ 0.74	11.47 $\pm$ 0.22	28.26 $\pm$ 0.95	81.96 $\pm$ 1.10	30.06 $\pm$ 0.82	34.17 $\pm$ 0.71	83.14 $\pm$ 0.93
Resp-SFT	42.18 $\pm$ 0.86	72.92 $\pm$ 0.98	11.62 $\pm$ 0.24	31.42 $\pm$ 1.35	83.88 $\pm$ 1.42	30.96 $\pm$ 0.97	35.04 $\pm$ 0.88	84.36 $\pm$ 1.21
CoT-SFT	41.93 $\pm$ 0.90	72.41 $\pm$ 1.02	12.06 $\pm$ 0.21	33.12 $\pm$ 1.28	83.54 $\pm$ 1.50	30.74 $\pm$ 0.95	34.92 $\pm$ 0.91	84.09 $\pm$ 1.18
Logit-SFT	42.57 $\pm$ 0.82	73.56 $\pm$ 0.95	11.81 $\pm$ 0.23	32.24 $\pm$ 1.22	84.06 $\pm$ 1.37	31.13 $\pm$ 0.93	35.21 $\pm$ 0.86	84.58 $\pm$ 1.16
Resp-KD	42.79 $\pm$ 0.80	74.03 $\pm$ 0.92	11.93 $\pm$ 0.22	32.68 $\pm$ 1.18	84.22 $\pm$ 1.30	31.42 $\pm$ 0.90	35.44 $\pm$ 0.83	84.92 $\pm$ 1.10
CoT-KD	42.44 $\pm$ 0.83	73.37 $\pm$ 0.94	12.31 $\pm$ 0.19	34.08 $\pm$ 1.12	83.91 $\pm$ 1.28	31.02 $\pm$ 0.88	35.29 $\pm$ 0.85	84.76 $\pm$ 1.08
Logit-KD	43.06 $\pm$ 0.78	74.74 $\pm$ 0.88	12.04 $\pm$ 0.21	33.07 $\pm$ 1.10	84.55 $\pm$ 1.22	31.77 $\pm$ 0.86	35.73 $\pm$ 0.80	85.27 $\pm$ 1.02
ReAD	44.63 $\pm$ 0.73	78.57 $\pm$ 0.86	12.86 $\pm$ 0.17	36.24 $\pm$ 1.05	86.38 $\pm$ 1.18	33.41 $\pm$ 0.79	36.98 $\pm$ 0.77	87.23 $\pm$ 0.95

Table 5: Target-specific distillation performance under a 150M-token budget (Qwen). ReAD outperforms all six one-hot baselines on every target.

Method	General	Steerability	Reasoning	Math	Code	Tool Use	LCU	Multilingual
Teacher only	55.27 $\pm$ 0.46	92.36 $\pm$ 0.52	14.28 $\pm$ 0.18	55.74 $\pm$ 0.78	90.12 $\pm$ 0.88	36.08 $\pm$ 0.63	40.23 $\pm$ 0.57	92.34 $\pm$ 0.66
Student only	40.07 $\pm$ 0.61	70.18 $\pm$ 0.74	11.47 $\pm$ 0.22	28.26 $\pm$ 0.95	81.96 $\pm$ 1.10	30.06 $\pm$ 0.82	34.17 $\pm$ 0.71	83.14 $\pm$ 0.93
Resp-SFT	48.02 $\pm$ 0.92	82.07 $\pm$ 1.04	13.18 $\pm$ 0.20	41.26 $\pm$ 1.38	86.92 $\pm$ 1.40	33.29 $\pm$ 0.96	37.42 $\pm$ 0.88	88.07 $\pm$ 1.20
CoT-SFT	48.29 $\pm$ 0.88	82.61 $\pm$ 0.98	13.72 $\pm$ 0.18	44.43 $\pm$ 1.30	86.58 $\pm$ 1.45	33.05 $\pm$ 0.94	37.28 $\pm$ 0.90	87.96 $\pm$ 1.18
Logit-SFT	48.46 $\pm$ 0.86	83.14 $\pm$ 0.95	13.39 $\pm$ 0.19	42.68 $\pm$ 1.26	87.05 $\pm$ 1.32	33.47 $\pm$ 0.92	37.63 $\pm$ 0.86	88.31 $\pm$ 1.12
Resp-KD	48.78 $\pm$ 0.84	83.76 $\pm$ 0.90	13.51 $\pm$ 0.18	43.62 $\pm$ 1.22	87.37 $\pm$ 1.26	33.92 $\pm$ 0.88	38.02 $\pm$ 0.82	88.64 $\pm$ 1.06
CoT-KD	49.05 $\pm$ 0.82	84.28 $\pm$ 0.88	13.87 $\pm$ 0.17	45.17 $\pm$ 1.18	87.11 $\pm$ 1.20	33.66 $\pm$ 0.86	37.91 $\pm$ 0.84	88.53 $\pm$ 1.02
Logit-KD	49.37 $\pm$ 0.80	85.19 $\pm$ 0.84	13.62 $\pm$ 0.18	44.07 $\pm$ 1.16	87.63 $\pm$ 1.18	34.26 $\pm$ 0.84	38.33 $\pm$ 0.78	88.92 $\pm$ 0.98
ReAD	51.53 $\pm$ 0.76	88.04 $\pm$ 0.90	14.03 $\pm$ 0.15	47.06 $\pm$ 1.05	89.04 $\pm$ 1.12	35.52 $\pm$ 0.78	39.47 $\pm$ 0.74	90.54 $\pm$ 0.92

Table 6 reports how much each method gains per additional 10M tokens between the two budgets, and Table 2 in the main text gives interpolation-based estimates of how many tokens the strongest one-hot baseline would need to reach ReAD’s 20M score on each of the eight capabilities.

Exact values. Figure 3 in Section 6 shows ReAD’s advantage in every comparison with its confidence interval. Table 7 lists the XSTest scores shown in Figure 5c, Table 8 the values of Figure 4, and Table 9 lists the differences and Welch t statistics shown in Figure 3, with significance marks after Holm correction within each block and budget, exactly as in the figure.

(The comparison with capability-specific distillation methods, the XSTest evaluation, and the significance analysis were moved to Section 6.)

F RELATED WORK

Capability Distillation for Large Language Models. Recent work on capability distillation has focused on transferring specific capabilities such as instruction following and context manipulation (Taori et al., 2023; Chiang et al., 2023; Peng et al., 2023; Xu et al., 2023a), thinking patterns (Zhang et al., 2024; Cheng et al., 2023), or text and code generation capability (Zhang et al., 2025; Li et al., 2023). These methods typically steer a teacher LLM via prompts or rationales, generate distillation data or logits, and fine-tune the student on that evidence. These methods typically devote the supervision budget to the target capability or domain itself. ReAD asks a complementary question: how to allocate a fixed budget across capability-targeted supervision so as to improve a specified target capability, rather than spending all of it on the target itself.

Distillation objectives, data mixtures, and forgetting. Beyond capability-specific recipes, LLM distillation methods differ in where the training sequences come from. Sequence-level KD and its descendants train on teacher-generated outputs (Hinton et al., 2015; Kim & Rush, 2016), whereas on-policy methods train on student-generated sequences scored by the teacher, which reduces exposure bias (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024). ReAD is off-policy in this sense and treats the objective as a plug-in: its contribution is the allocation of a fixed budget across capabilities, not a new loss. A separate line of work optimizes the *mixture* of training data: DoReMi (Xie et al., 2023) and RegMix (Liu et al., 2024) learn static domain weights before, or without observing, the target model’s training trajectory. Closest to ReAD are methods that adapt the mixture during training. Skill-It (Chen et al., 2023) estimates a graph of prerequisite relations among skills and

Table 6: Marginal gain from the 20M to the 150M budget, in points per additional 10M tokens, computed from Table 1 as $(s_{150M} - s_{20M})/13$. Bold: largest per column.

Method	General	Steer.	Reason.	Math	Code	Tool	LCU	Multi.	Mean
Resp-SFT	0.39	1.22	0.04	0.89	0.39	0.12	0.11	0.63	0.48
CoT-SFT	0.38	1.21	0.05	1.00	0.39	0.12	0.13	0.61	0.49
Logit-SFT	0.42	1.26	0.04	0.92	0.40	0.13	0.12	0.65	0.49
Resp-KD	0.45	1.34	0.05	0.95	0.45	0.15	0.14	0.67	0.52
CoT-KD	0.44	1.30	0.05	1.05	0.44	0.15	0.13	0.66	0.53
Logit-KD	0.47	1.42	0.05	1.01	0.46	0.16	0.15	0.70	0.55
ReAD	0.52	1.55	0.03	1.09	0.53	0.11	0.16	0.85	0.60

Table 7: Held-out XSTest evaluation. “Best baseline” denotes the best of the six SFT/KD baselines for each metric **and budget separately**.

Method	20M safe ↓	20M unsafe ↑	150M safe ↓	150M unsafe ↑
Best baseline	15.1 ± 0.77	75.5 ± 1.09	13.1 ± 0.69	79.0 ± 0.88
ReAD	13.8 ± 0.72	78.1 ± 0.97	11.8 ± 0.58	81.6 ± 0.83

updates skill-sampling weights online, including for a single target skill; ODM (Albalak et al., 2023) treats data domains as arms of a multi-armed bandit rewarded by the training loss; and Aioli (Chen et al., 2025) estimates the parameters of a mixing law during training and uses them to adjust the proportions. ReAD shares this online, feedback-driven view and studies target-oriented allocation in teacher-supervised distillation: supervision is generated on the fly by a teacher, the reward is built from measured changes in a capability profile rather than from the training loss, and regressions on prioritized capabilities receive an extra penalty. Finally, the off-target degradation we measure is related to catastrophic forgetting in continual fine-tuning (Kirkpatrick et al., 2017; Luo et al., 2023); ReAD adds no explicit anti-forgetting regularizer but penalizes measured regressions on prioritized capabilities directly in its reward (Eq. 6).

G BROADER IMPACT

ReAD aims to make capability distillation more budget-efficient by allocating limited training tokens across interdependent LLM capabilities. A positive impact is that smaller student models can become more useful under fixed compute budgets, potentially reducing deployment cost and improving access to LLM-based systems in resource-constrained settings. ReAD also explicitly monitors cross-capability transfer, which can help practitioners diagnose when improving one capability degrades other capabilities that a deployment depends on.

At the same time, more efficient distillation may lower the cost of producing specialized models with dual-use capabilities, such as code generation, tool use, or instruction following. Distilled models may also inherit biases, unsafe behaviors, privacy risks, or other limitations from teacher models and training data. In addition, if the capability probes or **capability-priority** vectors are misaligned with the intended deployment objective, ReAD may optimize benchmark utility while overlooking safety-, fairness-, or robustness-relevant behaviors.

We therefore view ReAD as a research framework rather than a deployment-ready safety guarantee. For sensitive applications, practitioners should combine ReAD with task-specific safety evaluation, robustness and bias audits, data filtering, red-team testing, and monitoring of off-target capability changes. Any release or deployment of distilled models should follow the usage restrictions of the underlying models and include appropriate documentation of intended use, limitations, and the safety evaluations that were actually performed.

H LIMITATION AND FUTURE WORK

While ReAD improves budget efficiency by using low-cost capability monitoring and adaptive allocation, it still relies on two practical components that can limit performance. First, the task requirement estimator and the probe suite may be imperfect: if the inferred **capability priorities** or probe signals are misaligned with true task utility, ReAD can allocate budget suboptimally. In

Table 8: Exact values behind Figure 4. (a) 150M-token gain–spillover trade-off from the capability transfer matrix. (b) Matched-budget allocation schedules under the same generator.

(a) Harmful transfer at 150M				(b) Allocation schedules		
Method	On-tgt $\uparrow$	Off-tgt $\Delta \uparrow$	Neg. $\downarrow$	Method	20M $\uparrow$	150M $\uparrow$
Single-cap. KD	4.79$\pm$0.29	-1.57 $\pm$ 0.21	0.43 $\pm$ 0.03	Uniform static	60.94 $\pm$ 0.48	64.73 $\pm$ 0.36
Task-static mix	4.61 $\pm$ 0.25	-0.87 $\pm$ 0.17	0.29 $\pm$ 0.03	Task-static mix	61.71 $\pm$ 0.43	65.52 $\pm$ 0.33
Greedy one-step	4.72 $\pm$ 0.23	-0.94 $\pm$ 0.19	0.31 $\pm$ 0.03	Greedy one-step	62.03 $\pm$ 0.41	66.01 $\pm$ 0.30
Grid-searched	4.66 $\pm$ 0.24	-0.71 $\pm$ 0.16	0.25 $\pm$ 0.03	Grid-searched	61.88 $\pm$ 0.42	65.81 $\pm$ 0.31
ReAD	4.79 $\pm$ 0.24	-0.38$\pm$0.12	0.17$\pm$0.02	ReAD	63.04$\pm$0.39	67.21$\pm$0.28

Table 9: Welch t -tests (three seeds, unpaired, two-sided) on the means and standard deviations of Tables 1, 8(b), and 7. Δ is ReAD minus the competitor; p values are Holm-corrected within each block and budget, and across all four tests for XSTest (*: $p < 0.05$, **: $p < 0.01$).

Comparison	Competitor	20M		Competitor	150M	
		Δ	t		Δ	t
General	Logit-KD	+1.00	2.9	Logit-KD	+1.62	4.1
Steerability	Logit-KD	+3.82	7.4*	Logit-KD	+5.57	9.1**
Reasoning	CoT-KD	+0.39	3.8	CoT-KD	+0.14	1.6
Math	CoT-KD	+1.72	3.0	CoT-KD	+2.18	3.1
Code	Logit-KD	+2.44	2.9	Logit-KD	+3.33	3.8
Tool Use	Logit-KD	+2.41	4.4	Logit-KD	+1.82	3.6
LCU	Logit-KD	+1.86	4.6	Logit-KD	+1.87	3.8
Multilingual	Logit-KD	+2.23	3.1	Logit-KD	+4.13	5.4*
Avg. utility (Table 8b)	Greedy one-step	+1.01	3.1	Greedy one-step	+1.20	5.1**
Avg. utility (Table 8b)	Grid-searched	+1.16	3.5	Grid-searched	+1.40	5.8**
Avg. utility (Table 8b)	Task-static mix	+1.33	4.0	Task-static mix	+1.69	6.8**
Avg. utility (Table 8b)	Uniform static	+2.10	5.9*	Uniform static	+2.48	9.4**
XSTest safe-refusal $\downarrow$	best baseline	-1.3	-2.1	best baseline	-1.3	-2.5
XSTest unsafe-refusal $\uparrow$	best baseline	+2.6	3.1	best baseline	+2.6	3.7

particular, the proxy reward is computed on a small probe suite and may deviate from the true task utility; we do not report a quantitative validation of this proxy, so errors in it can mislead the allocation. Second, our capability set and data generator are necessarily finite and template-driven, which may not cover all behaviors a real task depends on, especially for highly domain-specific or long-horizon tasks. Future work can address these limitations by learning more robust and transferable task requirement models, designing probe suites with stronger coverage and better calibration to task utility, and expanding capability definitions and data generation beyond templates (e.g., automatic discovery of capability axes and open-ended curriculum generation). Another promising direction is to jointly optimize allocation and student training dynamics (e.g., optimizer schedules and parameter-efficient adapters) under the same budget, and to extend ReAD to settings with distribution shift, multi-task deployment, or continual updates where requirements and transfer patterns evolve over time.

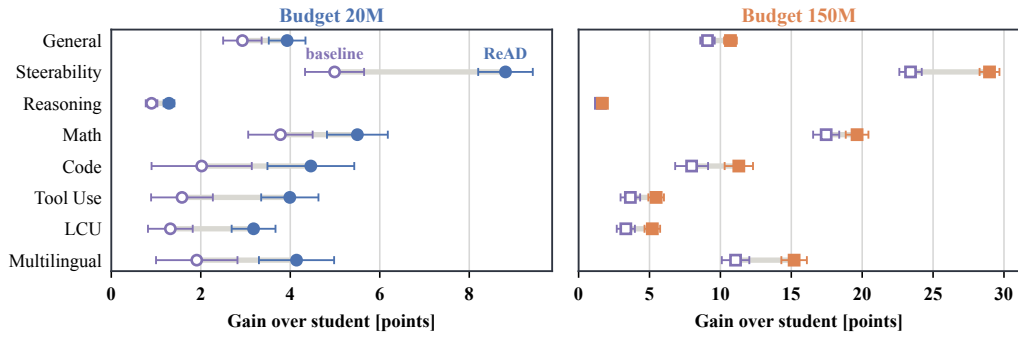


Figure 7: Points gained over the initial student on each capability at 20M (left) and 150M (right) for ReAD and for the best one-hot baseline of that capability, read off Table 1. Error bars are the standard deviation over three seeds of the distilled student’s score. Values are absolute percentage points, so small changes stay small.